

GOPI CHAITANYA ALAMANDA

Railway Technology Engineer | Python Automation & System Integration

+91 95422 30366 | gopichaitanyaalamanda4935@gmail.com | Open to Relocate

[Gopi Chaitanya Alamanda](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Railway Technology Professional with around 3 Years of experience at Efftronics Systems Pvt. Ltd. specializing in railway signaling application deployment, core technical support, real-time data analytics, and Python-based automation. Proven track record of optimizing mission-critical engineering workflows across 1,000+ Indian Railways stations. Strong grasp of signaling principles, interlocking concepts, and safety-critical system logic. Equipped with deep foundational/theoretical knowledge of modern train control systems (CBTC, Kavach)

CORE TECHNICAL & RAILWAY DOMAIN SKILLS

- **Railway Systems (Deployment & Support):** Train Monitoring Systems (TMS), Automatic Train Tracking Systems (ATTS), Remote Diagnostic and Preventive Maintenance (RDPM), Safety Point Alarm Unit (SPAU), Control Office Application (COA), Nirdesh Alerts.
- **Signaling & Interlocking Concepts:** Route Setting, Route Locking, Route Releasing & Route Cancellation; Route Relay Interlocking (RRI) & Panel Interlocking (PI) Station Operations.
- **Block Section Working Knowledge:** Absolute Block System (ABS), Automatic Signaling, Intermediate Block Huts (IBH).
- **Advanced Control (Theoretical):** Kavach (Indian ATP), ATO, ATS.
- **Software & Automation Engineering:** Python (Expert), Oracle SQL (Expert), Tkinter, Flask basic, Django basic, Azure VM, JSON Processing, Advanced Excel Automation, Pandas basic, NumPy basic, TCP/IP, IoT Monitoring, Git, Linux.

PROFESSIONAL EXPERIENCE

Efftronics Systems Pvt. Ltd. | Engineer – Technical Support & Software Automation

June 2023 – Present | Vijayawada, Andhra Pradesh

- Fault/Alarm/Failure Analysis to Find the Root cause of Failure.
- Checking/Analyzing Train Movements in case of Failures.
- Hands on Experience on Interlocking concepts of Indian Railways.
- Configured, deployed, and stabilized high-availability railway signaling and train tracking applications for 1,000+ railway stations.
- Designed and engineered Digital Twin visualization systems for critical signaling circuits (Signals, Points, Track Circuits, Battery Health Monitoring Units) to execute real-time flow and health monitoring.
- Formulated advanced Oracle SQL analytical queries to identify train tracking break patterns on ATTS data, directly resulting in tracking algorithm upgrades in collaboration with CRIS.
- Built python based automated testing utilities to generate route-wise validation test cases for occupied platform conditions (SPAU), lowering validation cycle times across 300+ station.
- Reduced 30 Days of manual work by Developing python application which automates the process of doing 30 Days of manual work into 5 minutes.
- Hand-picked to lead a team of 7 interns & 5 employees tasked with deploying the Nirdesh Alerts real-time failure module across 145 target stations in Jaipur, Allahabad & Jhansi divisions.
- Lead & Contributed to the system deployment of India's largest TMS [Train Monitoring Systems] inaugurated by Railway Minister.

- Selected to present live technical solutions at the IREE and IRSE exhibitions in New Delhi.
- Prepared and presented rigorous Acceptance Test Procedures (ATPs) to RDSO and RITES officials to facilitate official system validation and product sign-off.
- Providing 24*7 L2 Technical Support to Service Engineers & Railway Clients
- Troubleshoot Software/System Architectural related issues

MAJOR IMPACT PROJECTS

Fault Standardization Automation Tool (Fault Replacer) *(Python, Oracle SQL)*

Problem: 18 Eastern Dedicated Freight Corridor (EDFCC) projects required manual standardization of alarm/fault names across 61 stations—a workflow estimated at 30 days of manual effort.

Solution: Built a Python processing script with intelligent before/after data mapping to execute bulk database modifications automatically.

Impact: Reduced execution time from 30 days to under 5 minutes with 100% data accuracy and zero visual config discrepancies.

ChronoTech (Database Skeleton Prep Utility) *(Python, Oracle, Excel/CSV)*

Problem: Manual configuration of database schemas from bulk data feeds required 15 minutes per 4,096 station asset records.

Solution: Programmed an automatic parsing engine that generated structural database dumps dynamically from source sheets.

Impact: Compressed processing time from 15 minutes to 2 minutes, earning an immediate corporate transition from site operations to the core engineering division.

Engineering Automation Suite *(Python, JSON, TCP/IP, File Systems)*

Developed a 4-tool workflow network used internally across multiple deployment branches:

- ESPAU Test Generator: Automated route safety test suites, scaling validation capacity across 300+ stations (90 min to 30 min per station).
- JSON Config Processor: Batch-edited asset properties, text labels, and flow orientations for Nirdesh UI dashboards instantly.
- Auto File Collector & Dump Creator: Automated localized Oracle backups and collects Application Config files for Cloud Deployment ready setups, eliminating hours of manual retrieval from nested date folders.

EDUCATION

BTech in Computer Science Engineering [AI & DS] | Kakinada Institute of Engineering and Technology
2025 – 2028 [Expected]

Diploma in Computer Science Engineering | Government Polytechnic College, Srikakulam, AP

Pursued: 2020 - 2023 | Aggregate: 80%

Secondary School Certificate [10th Class] | Government High School, Bheemunipatnam, AP

Pursued: 2020 | Aggregate: 80%

RECOGNITION & PROFESSIONAL CONTRIBUTIONS

Director Public Recognition: Formally cited by the corporate Director during an all-hands assembly for exceptional code quality and efficiency contributions.

- Executive Appreciation: Recipient of two official management letters of appreciation for pioneering engineering automation tools.
- Industry Representations: Selected as a core corporate technical presenter at the IREE 2023 and IRSE 2023 signaling and telecommunication exhibitions in New Delhi.